

Question	Answer	Mark	Mark scheme	Additional guidance
17 (a)	25.9	P1	for process to find volume of hemisphere, eg $\frac{1}{2} \times \frac{4}{3} \times \pi \times 3.5^3$ (=89.797...) $\left(\frac{343\pi}{12}\right)$ or for a correct expression for the volume of the cone, eg $\frac{1}{3} \times \pi \times 3.5^2 (y - 3.5)$ or $\frac{1}{3} \times \pi \times 3.5^2 \times h$	‘y – 3.5’ may be seen as a new variable, but cannot be just y Condone missing brackets Accept decimals rounded or truncated to 1dp 120π - "89.797..." = 287.193... or $\frac{1097\pi}{12}$ π may be missing throughout Award of this mark implies award of the previous May be seen in multiple steps
		P1	for setting up an equation linking all three aspects, eg $\frac{1}{2} \times \frac{4}{3} \times \pi \times 3.5^3 + \frac{1}{3} \times \pi \times 3.5^2 (y - 3.5) = 120\pi$ or “89.797...” + “12.828...”(y – 3.5) = “376.99 ...” or “28.5833...” π + “4.0833...” π(y – 3.5) = 120π	
		P1	for process to isolate y or (y – 3.5) or h in their equation, $120\pi - \frac{1}{2} \times \frac{4}{3} \pi 3.5^3 + \frac{1}{3} \pi 3.5^3$ eg $\frac{\frac{1}{3} \pi 3.5^2}{\frac{1}{3} \pi 3.5^2}$ or $\frac{"376.99..." - "89.797..." + "44.898..."}{"12.828..."}$ or $\frac{120\pi - "28.583..." \pi + "14.291..." \pi}{"4.083..." \pi}$ oe	
		A1	for answer in range 25.8 to 26.3 SCB3 for an answer in the range 22.3 to 22.8 or $\frac{1097}{49}$	
				If an answer is given in the range in working and then rounded incorrectly award full marks.

Question	Answer	Mark	Mark scheme
(b)	explanation	C1	for explanation, eg Acceptable examples the height would decrease the height would be 0 at 14.227 y would be smaller it would decrease Not acceptable examples the height would increase